Nathan Walker, BS Electrical Engineering

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Executive Summary

Master's-level Electrical Engineer (graduating May 2026) with strong experience in FPGA-based system design, digital signal processing, and embedded systems. I conducted graduate research in near-field array processing for sensing and noise cancellation applications, including real-time implementation of array processing and adaptive filtering algorithms. Revitalized a legacy research system with minimal documentation, transforming it into a fully operational and well-documented platform, and designed supporting hardware to expand system capability and usability. Experienced in implementing DSP algorithms on FPGA and DSP platforms, with strengths in system-level integration, hardware–software co-design, and translating complex theory into reliable, working systems.

Work Experience

Graduate Researcher May 2024 – Present

Thesis: *Design and Implementation of Adaptive Active Snore Reduction Using Near-Field Array Processing*

- Developed real-time embedded signal processing algorithms on Linux, implementing adaptive filtering and array processing for noise cancellation.
- Repaired and debugged custom electronics designed by others with no circuit documentation, identifying faults and restoring full functionality.
- Used MATLAB and C to analyze data, supporting debugging and performance optimization.
- Created custom PCB enclosures in Solid Edge to support mechanical stability and system function
- Effectively communicated complex technical problems and solutions to advising professors.
- Ensured data integrity during collection and analysis of automated experimental measurements.

ASPIRE Research Assistant May 2023 – April 2024

- Selected for the ASPIRE research program based on academic performance and faculty recommendation.
- Designed and executed PCB and component testing protocols, including soldering, signal verification, and measurement with lab instruments, ensuring accurate experimental results.
- Developed hardware solutions for research projects under size, cost, and performance constraints.
- Assembled complex PCBs and coordinated multi-component research projects from design to execution, demonstrating hands-on troubleshooting and project management.

Engineering Projects

International Future Energy Challenge – Team Lead July 2023 – July 2024

- Led a team of 3 undergraduates, taking primary responsibility for hardware design and component selection while delegating tasks to deliver a high-efficiency Class D amplifier.
- Designed and implemented custom magnetic components using Ansys simulations to meet performance goals.
- Created and assembled multiple PCBs using Altium, improving product quality and testability.
- Troubleshot and debugged custom PCBs under tight design constraints, applying circuit analysis, measurement techniques, and hands-on problem-solving.

Skills

Altium/KiCad	C/C++ & Python	MATLAB/Data Analysis
Linux/GitHub/Scripting	Solid Edge/ Mechanical Design	Signal Processing
Communication & Teamwork	Hardware Troubleshooting	VHDL/Verilog

Education

Utah State University – Logan Utah

- M.S. Electrical Engineering – 3.83 GPA May 2026
- B.S. Electrical Engineering – 3.97 GPA May 2025